

INSTITUTO TECNOLÓGICO AUTÓNOMO DE MÉXICO

ECONOMICS IV

ECO-21104

Credits

6.0 (3 hrs lecture, 1.5 hr laboratory)

Coordinator

Diego Domínguez Larrea

Textbook

- Osborne, Martin J. An Introduction to Game Theory, Oxford University Press, 2004.
- Gibbons, Robert Game Theory for Applied Economists, Princeton University Press, 1992.
- Varian, Hal. Microeconomía Intermedia, 4th edition, Antoni Bosch, 1996.
- Varian, Hal. Microeconomic Analysis, 3th edition, W.W. Norton & Company, 1992.
- Departamental notes made by the coordinator, Diego Domínguez Larrea.
- Hayek, F. "The Use of Knowledge in Society", American Economic Review, 35(4), 519-530.
- Charles A. Holt, Classroom Games: Trading in a Pit Market", The Journal of Economic Perspectives 10(1), 1996, 193-203.
- Jacob K. Goeree and Charles A. Holt, "Ten Little Treasures of Game Theory and Ten Intuitive Contradictions" mimeo, 2016.

Course description

Provide the student the theoretical tools used for general equilibrium competitive model. Introduction to the game theory analysis.

Pre-requisites

ECO-21103 Economics III and MAT-12101 Calculus II

Major status

Required

Assessment summary

Assessment task	Value	Linked learning outcomes
First departmental exam	30%	Topics: I-III
Second departmental exam	30%	Topics: III-IV
Final exam	40%	Topics: I-IV

Course topics

- I. General Equilibrium
 - A. Economies without production
 - Edgeworth box
 - Equilibrium
 - Efficiency
 - Welfare theorems
 - B. Economies with production
 - Equilibrium
 - Efficiency
 - Welfare theorems
 - Robinson Crusoe economy

- II. Monopoly and monopsony
 - A. Barriers to entry
 - B. Simple monopoly
 - C. First degree discrimination
 - D. Second degree discrimination
 - E. Third degree discrimination
 - F. Monopsony

- III. Game Theory
 - A. Normal form games (full information)
 1. Defining a game
 2. Examples (Prisoner's Dilemma, Coordination Games, Sexes battle, ...)
 3. Strict Dominance and Diligent Dominance
 4. Iterate Elimination of Dominated Strategies (Strict and Flexible)
 5. Mixed Strategies
 6. Best Answers
 7. Nash equilibrium
 8. Examples (Cournot, Bertrand, Voting, Medium Voter Theorem)
 - B. Games in extensive form (complete information)
 1. Definition

2. Strategies and Representation in Normal Form of an extensive form game
 3. Subgames
 4. Perfect Nash Balance in Subgames
 5. Examples (Stackelberg, Entry Deterrence)
- C. Repeated games
1. Stage Game
 2. Balance in the Stage Game
 3. Strategies in repeated games
 4. Folk Theorems

IV. Externalities

- A. Definition and examples
 1. Equilibrium and efficiency
 2. Solutions
 3. The Coase conjecture
- B. Network externalities
- C. Externalities of information
- D. Public goods